MD ABUL BASHAR

TechCorp Branch Office Active Directory & Network Infrastructure Setup

Project Phases: AD DS, DNS, DHCP, GPO, IPAM
Branch Domain: techcorp.local
Target Subnet: 192.168.10.0/24

Phase 1: The Foundation (Active Directory & DNS)

Objective & Mechanism

The objective is to establish the core identity and name resolution infrastructure.
Active Directory Domain Services (AD DS) provides centralized user authentication and policy management, while DNS enables domain name resolution for internal resources.

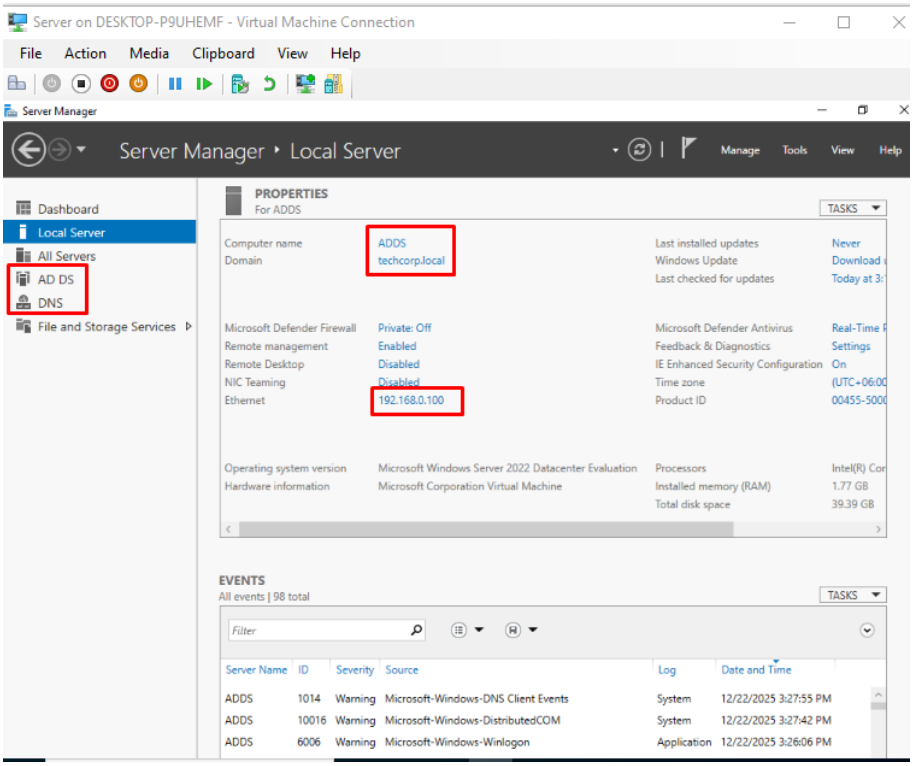
Implementation Steps

1. **Installed AD DS Role**

- Opened Server Manager → Add Roles and Features.
- Selected Active Directory Domain Services and completed installation.

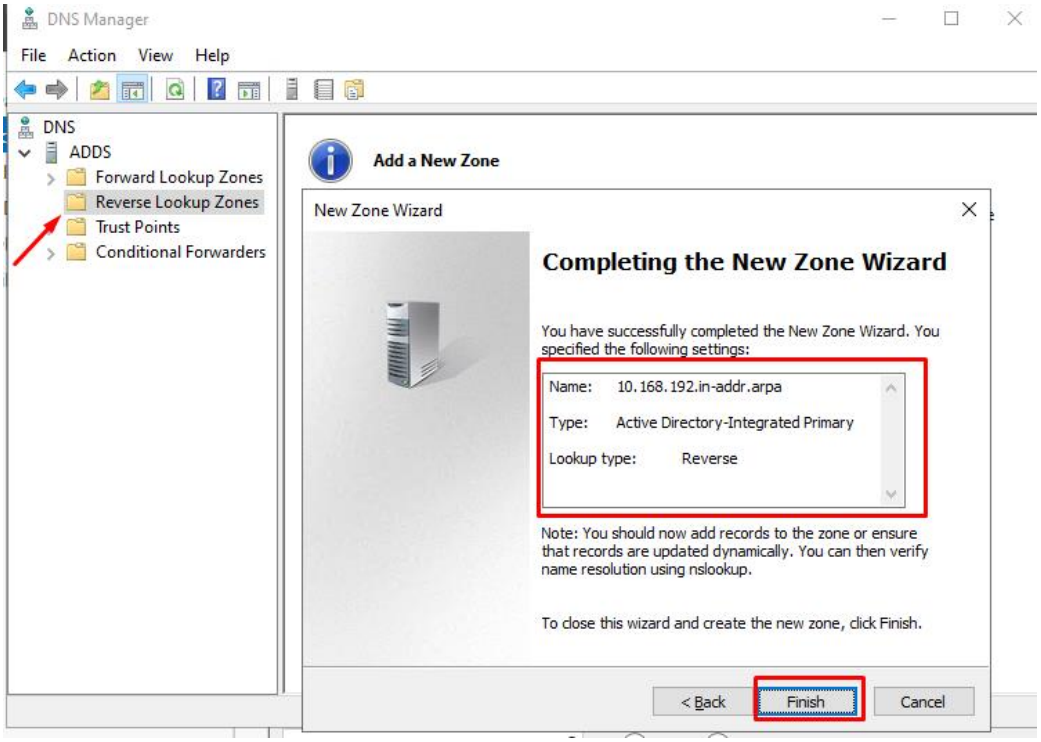
2.Promoted Server to Domain Controller

- From the notification flag, selected "Promote this server to a domain controller".
- Choose "Add a new forest" and entered the root domain name: techcorp.local.
- Set DSRM password and allowed the server to restart after promotion.



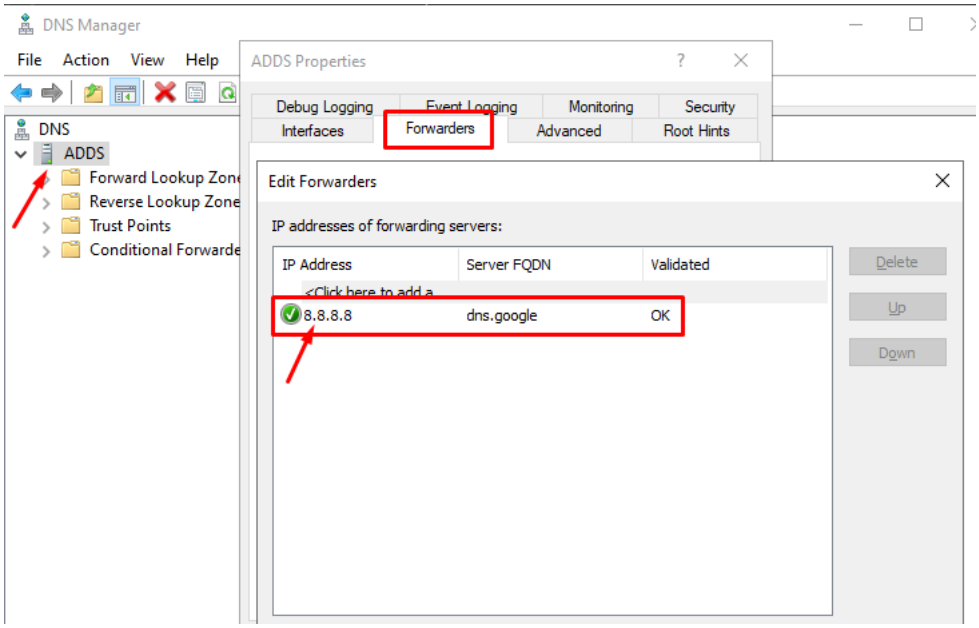
2. **Configured DNS Zones**

- Opened DNS Manager.
- Forward Lookup Zone for techcorp.local was created automatically.
- Created a **Reverse Lookup Zone** for subnet 192.168.10./24 to enable IP-to-hostname resolution.

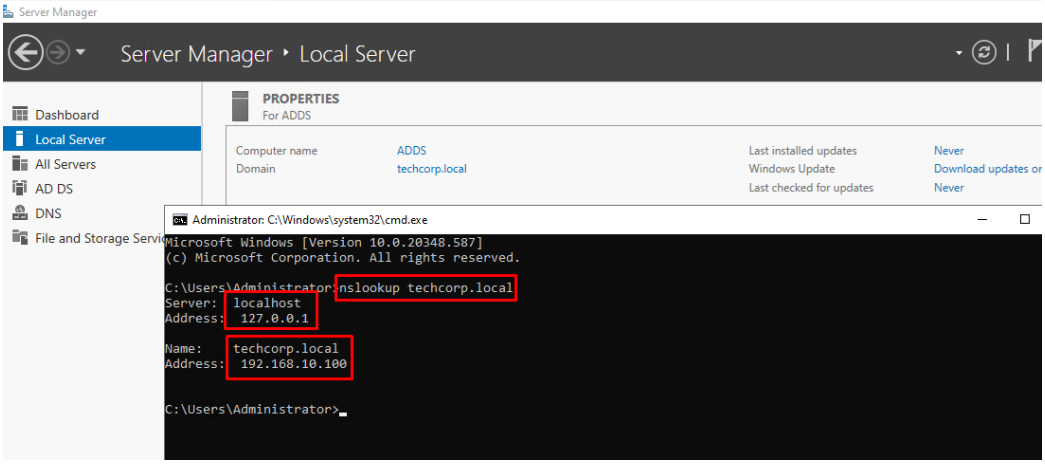


3. **Configured DNS Forwarder**

- In **DNS Manager**, right-clicked the server → **Properties** → **Forwarders**.
- Added 8.8.8.8 (Google DNS) to allow external DNS resolution for internal clients.



Success Verification



Expected Output: Returns the IP address of the DC (192.168.10.100).

Phase 2: Dynamic Networking (DHCP)

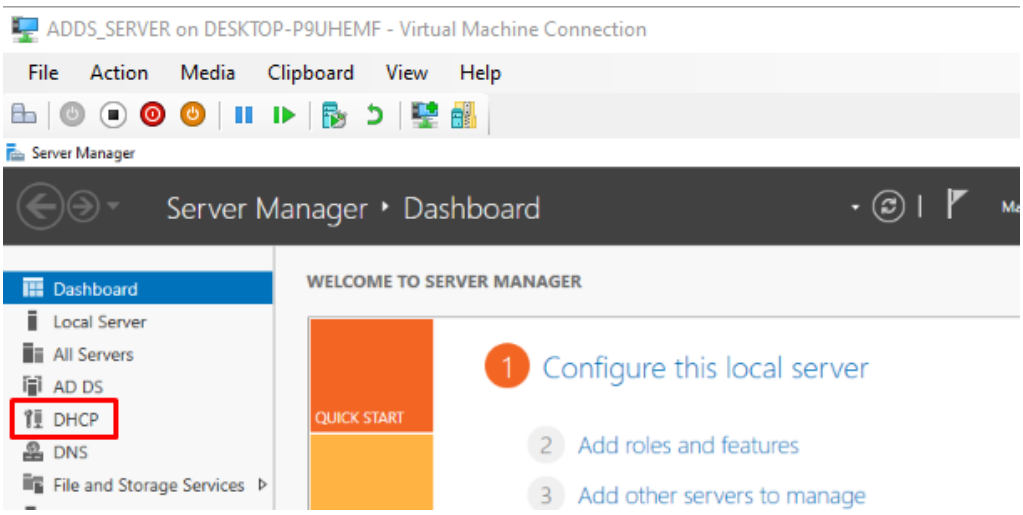
Objective & Mechanism

The objective is to automate IP address assignment and reduce manual configuration. DHCP dynamically assigns IPs and network settings (gateway, DNS) to clients, improving scalability and reducing errors.

Implementation Steps

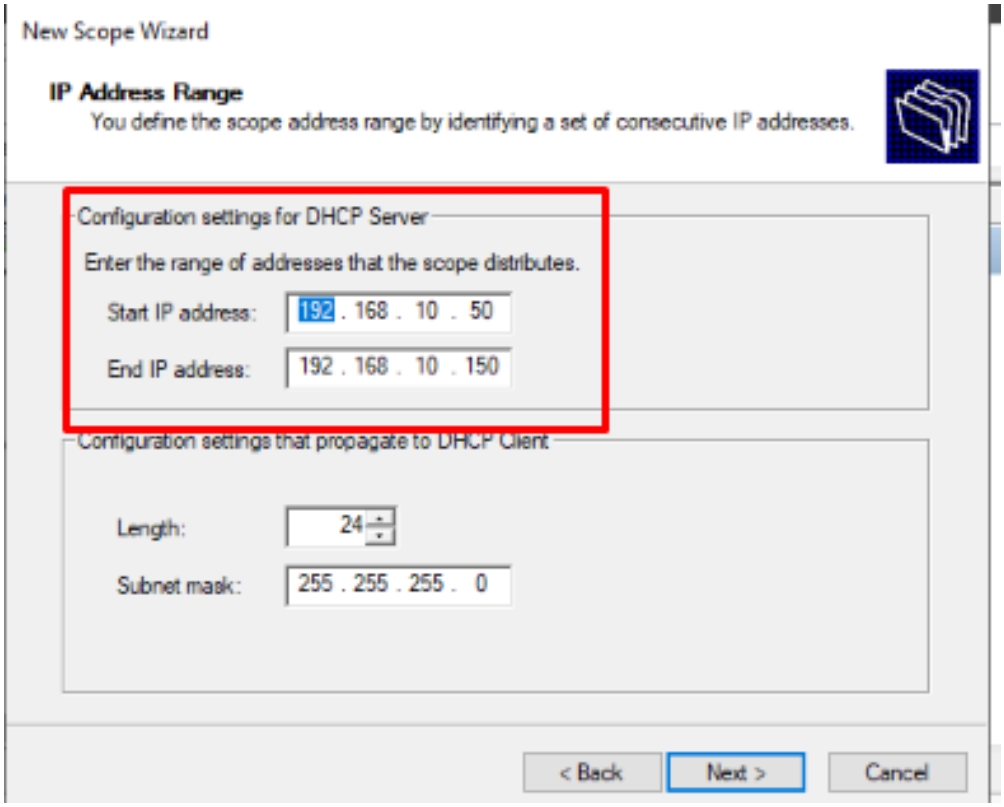
1. **Installed DHCP Role**

- Added **DHCP Server** role via Server Manager.



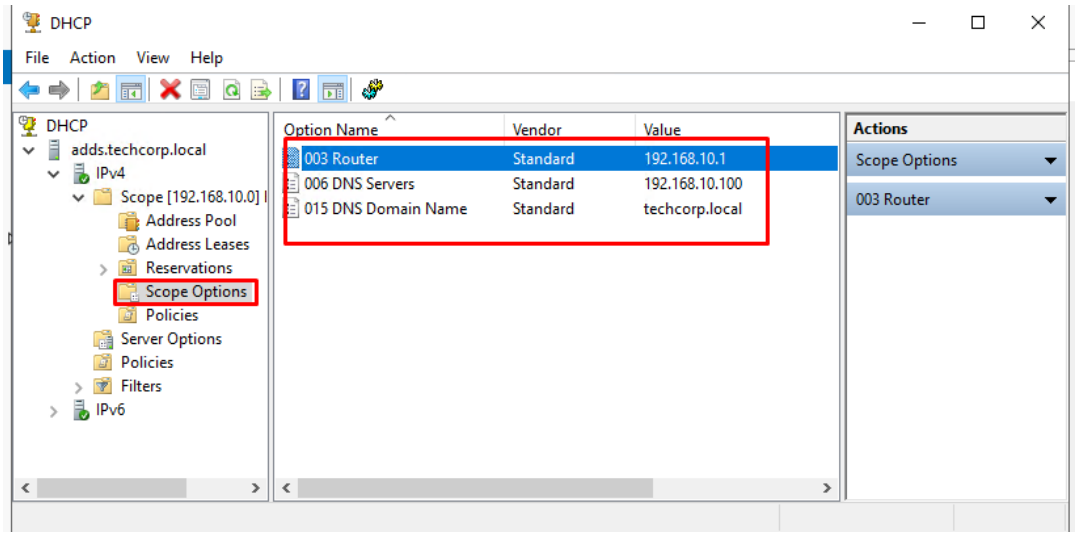
2. **Created DHCP Scope**

- Scope Name: Branch_Office_Scope
- IP Range: 192.168.10.50 – 192.168.10.150
- Subnet Mask: 255.255.255.0



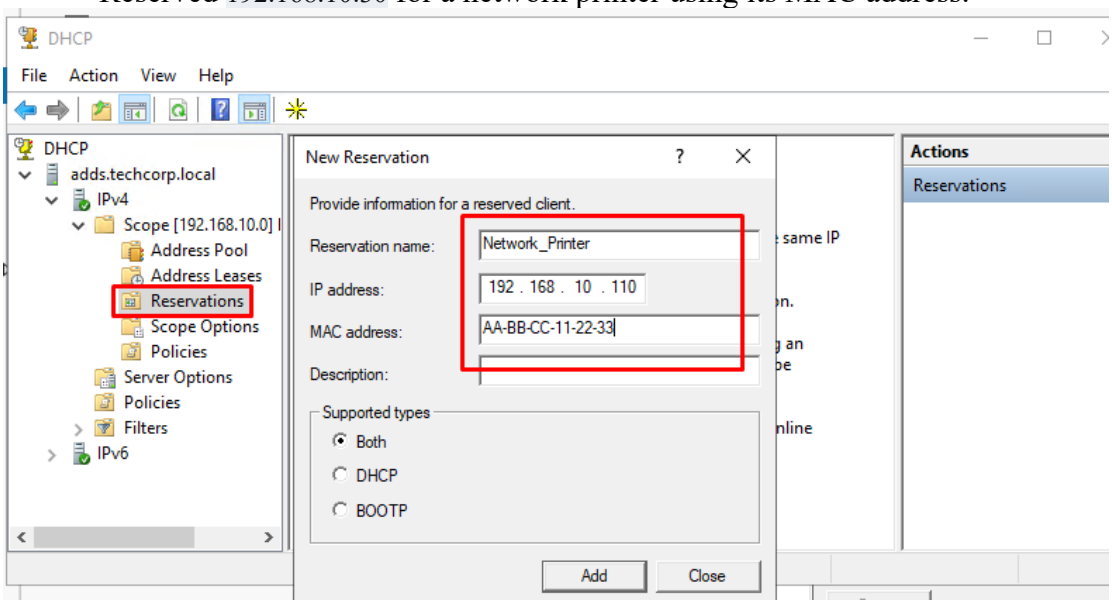
3. **Configured DHCP Options**

- **003 Router:** 192.168.10.1 (Default Gateway)
- **006 DNS Servers:** 192.168.10.100 (DC IP)
- **015 DNS Domain Name:** techcorp.local



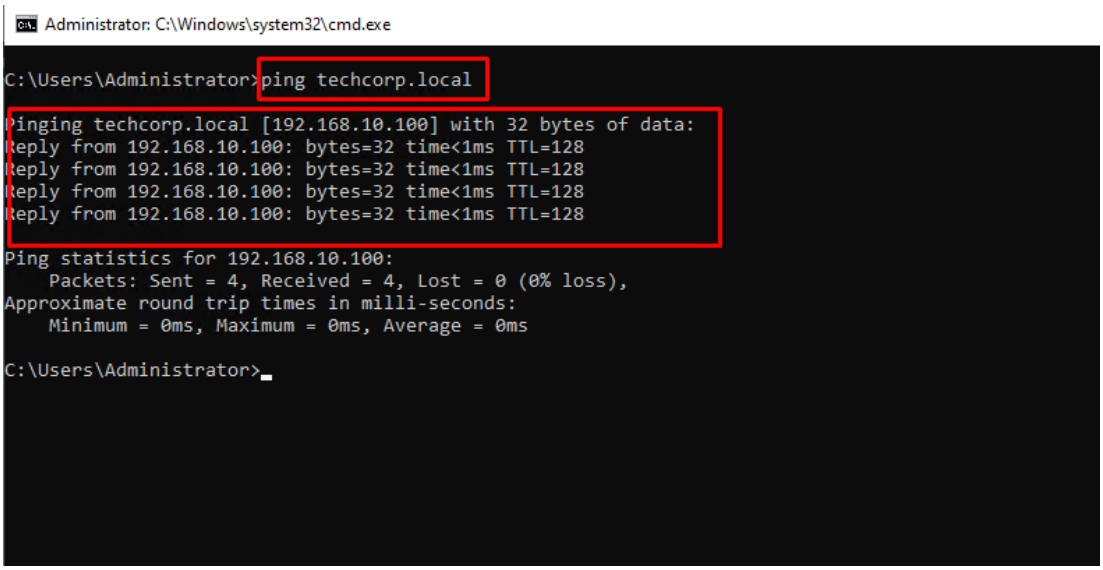
Created DHCP Reservation

- Reserved 192.168.10.30 for a network printer using its MAC address.



Success Verification

4. Client configured for DHCP receives an IP in the defined range.
5. Client can ping the DC by name:



Phase 3: Governance (Group Policy Objects - GPO)

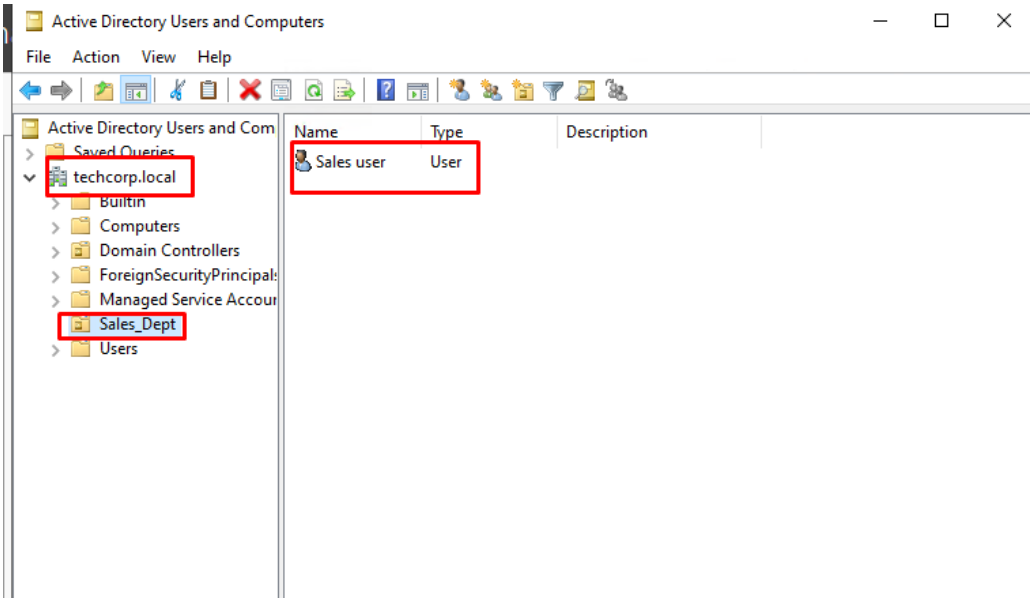
Objective & Mechanism

The objective is to enforce security and operational policies through centralized Group Policy. GPOs allow administrators to control user/computer settings across the domain, ensuring compliance and standardization.

Implementation Steps

1. **Created Organizational Unit (OU)**

- In **Active Directory Users and Computers**, created an OU named Sales_Dept.
- Moved a test user (sales_user) into the OU.

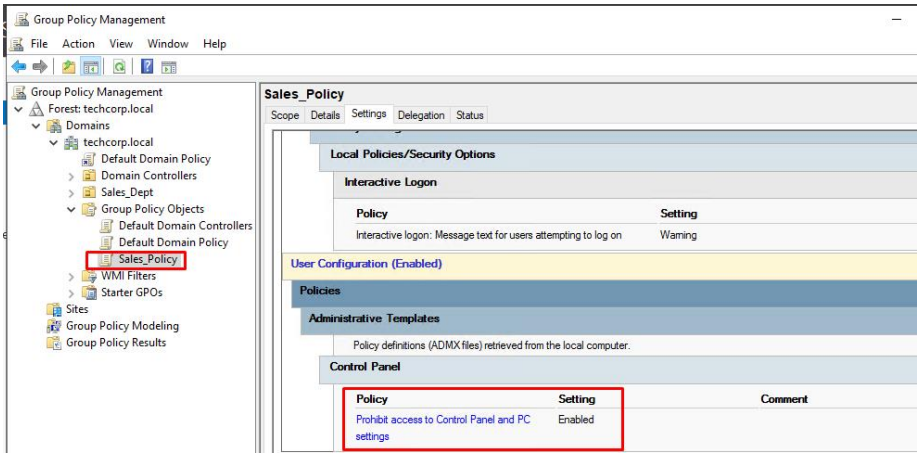


2. **Created and Linked GPO**

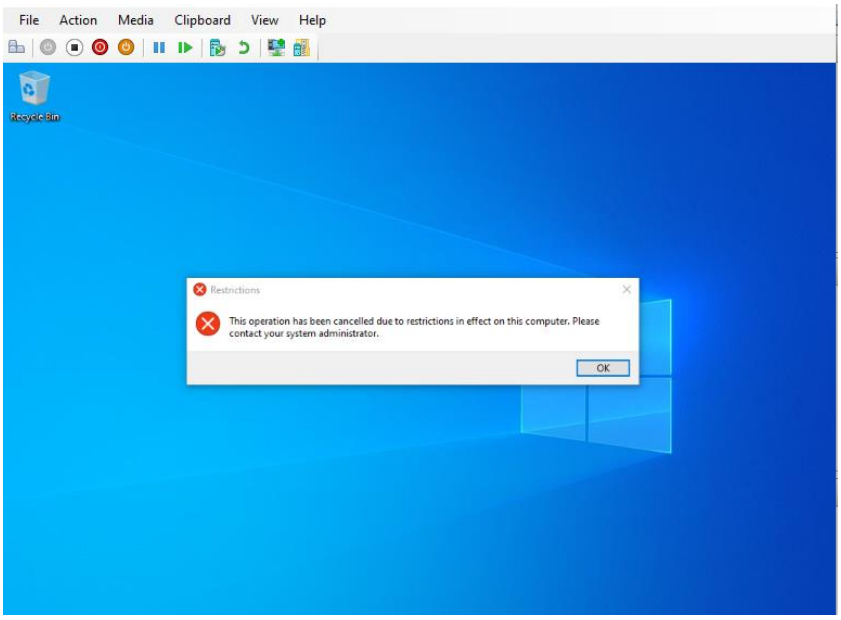
- Created a GPO named Sales Dept Policy and linked it to the Sales Dept OU.
- Configured the following policies:

3. a. **Control Panel Restriction**

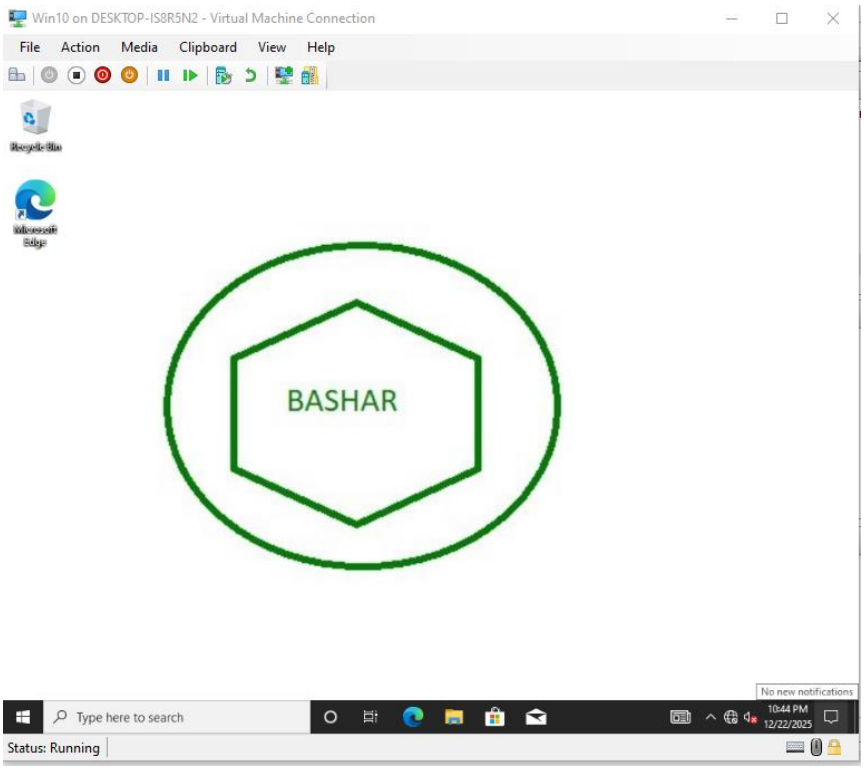
   - Press **Windows Key + R**, type gpedit.msc and hit **Enter or**
     Path: User Configuration → Policies → Administrative Templates → Control Panel
     Policy: **"Prohibit access to Control Panel and PC settings" → Enabled**.



Result:



The desktop wallpaper changed after I restarted the computer.



4. b. **Network Drive Mapping**
   Path: User Configuration → Preferences → Windows Settings → Drive Maps
   Action: **Update**
   Location: \\ADDS\Shared
   Drive Letter: S:

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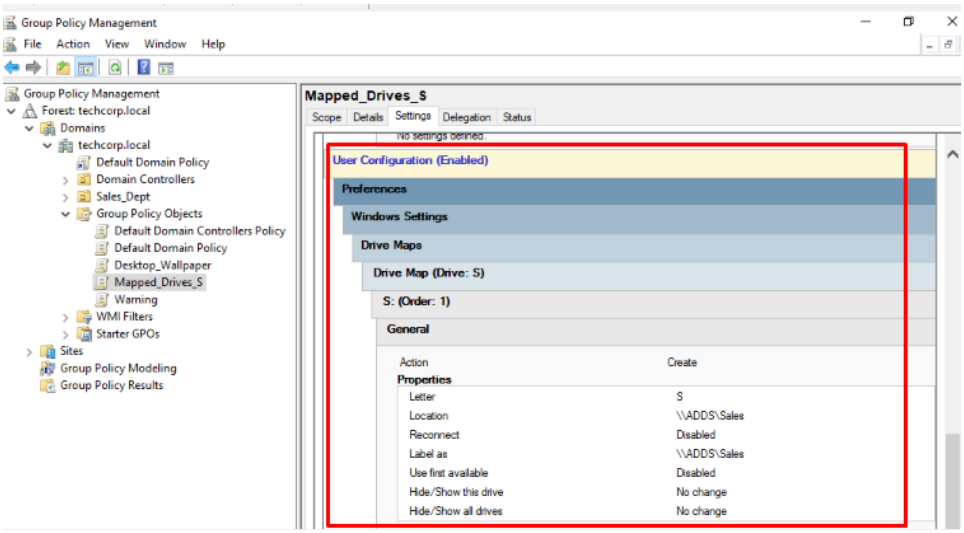
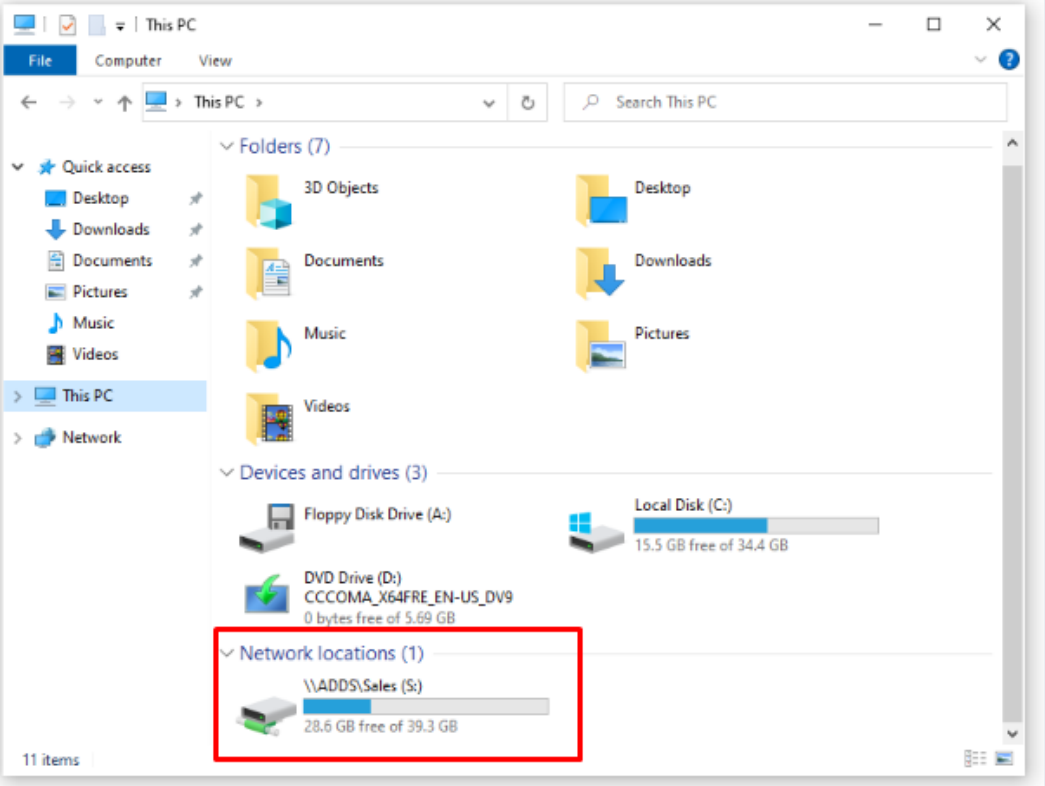
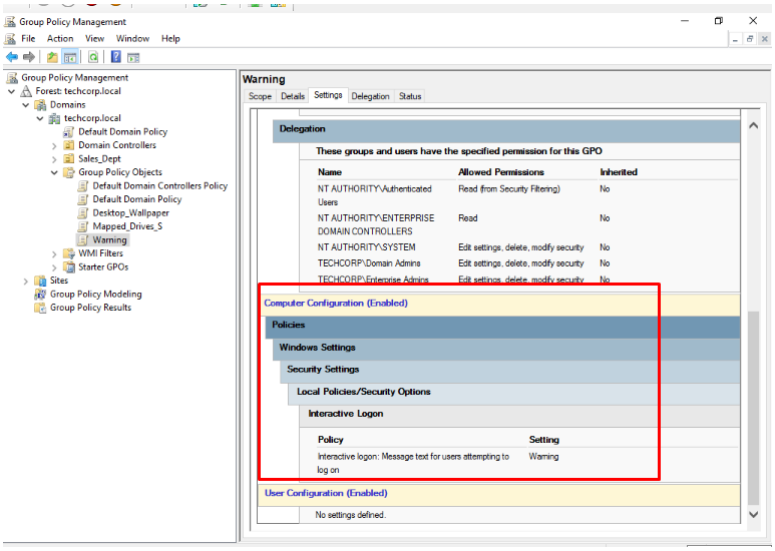


Result:



5. c. **Interactive Logon Message**

- Press **Windows Key + R**, type gpedit.msc, and hit **Enter or**
- Path: Computer Configuration → Policies → Windows Settings → Security Settings → Local Policies → Security Options
  Policy: **"Interactive logon: Message text for users attempting to log on"**
  Example text: *"Warning"*



Success Verification

- Logged in as sales_user.
- Verified that Control Panel is inaccessible.
- Confirmed the S: drive is mapped.
- Legal warning appears at log

Phase 4: Management & Monitoring (IPAM)

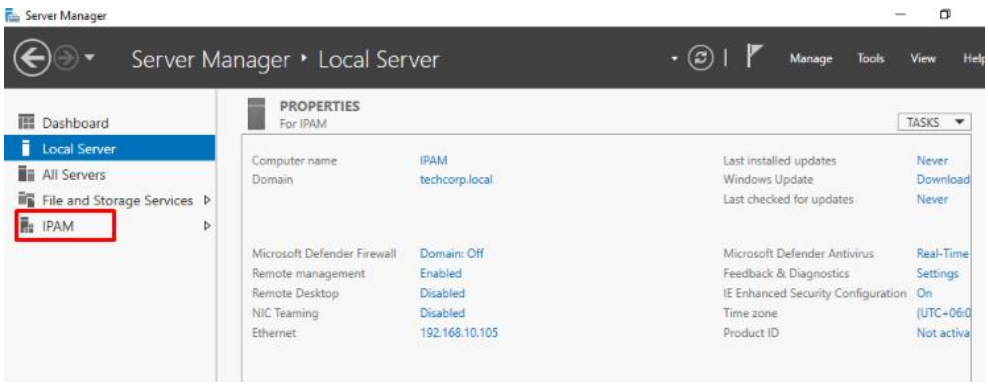
Objective & Mechanism

The objective is to centralize IP address management and monitoring.
IPAM provides a unified console for tracking IP usage, managing DHCP and DNS, and auditing IP address changes.

Implementation Steps

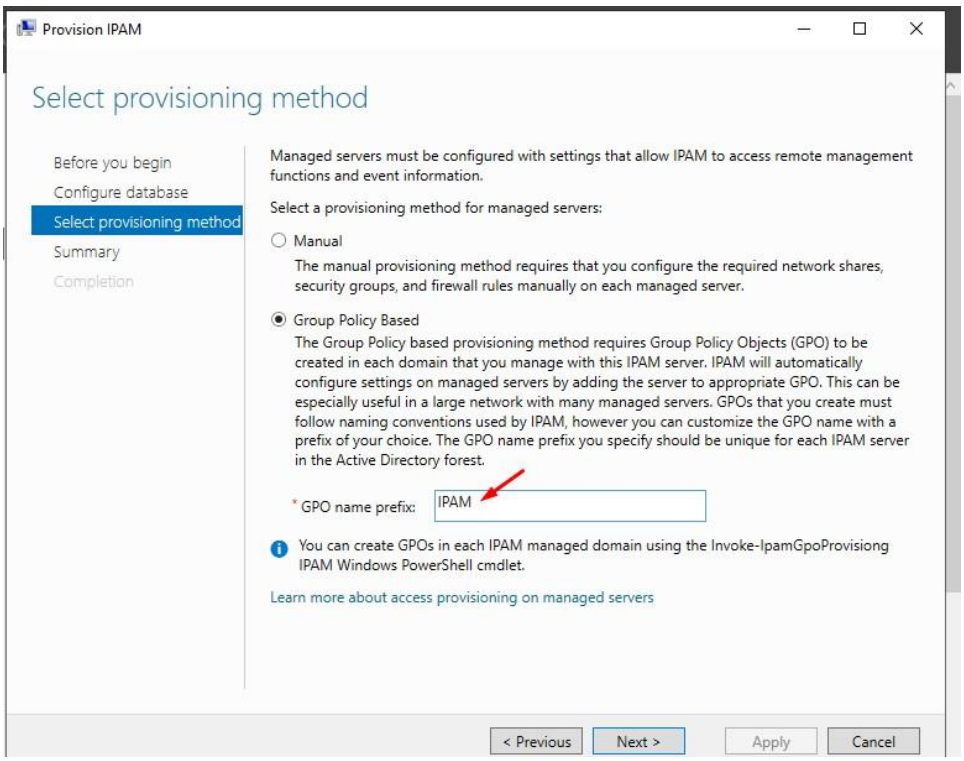
1. **Installed IPAM Role**

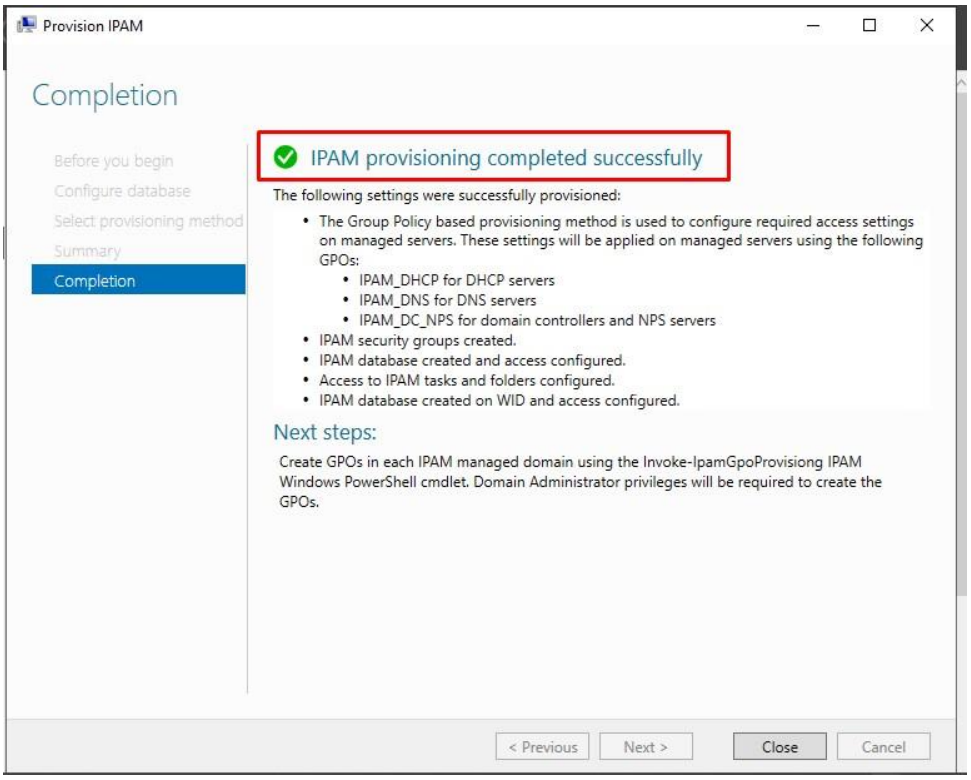
Installed **IP Address Management (IPAM)** on a member server (or DC in lab).



2. **Ran Provisioning Wizard**

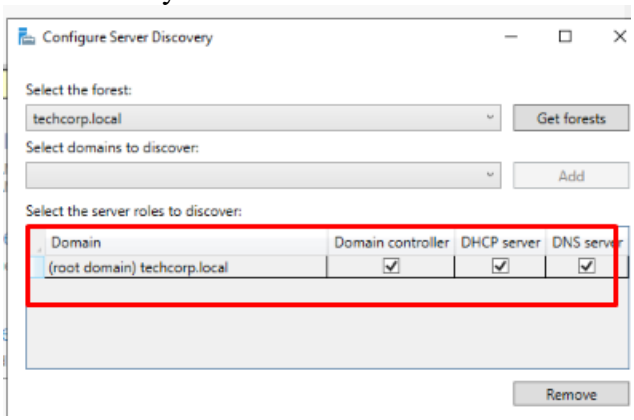
- Chose "GPO-based" provisioning.
- IPAM created GPOs (IPAM_DHCP, IPAM_DNS, IPAM_DC_NPS) to manage server access.





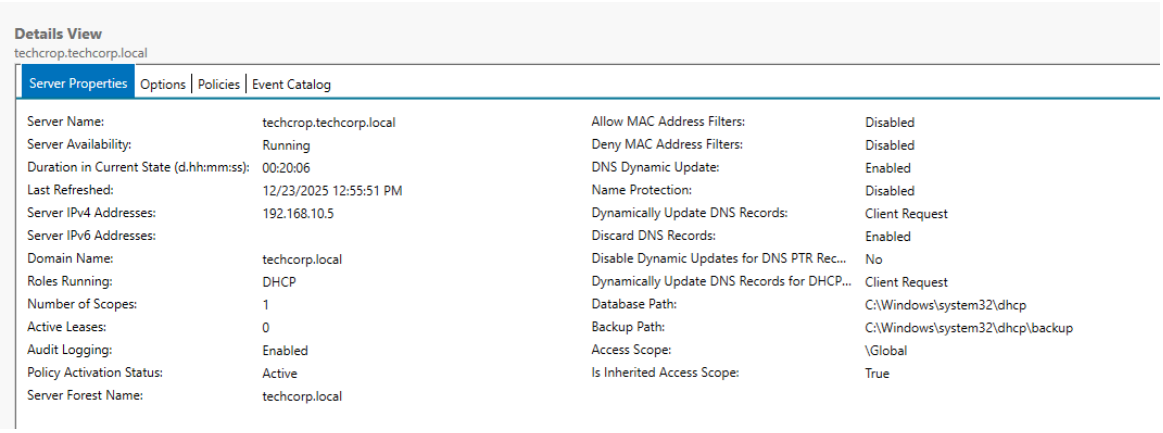
3. **Ran Server Discovery**

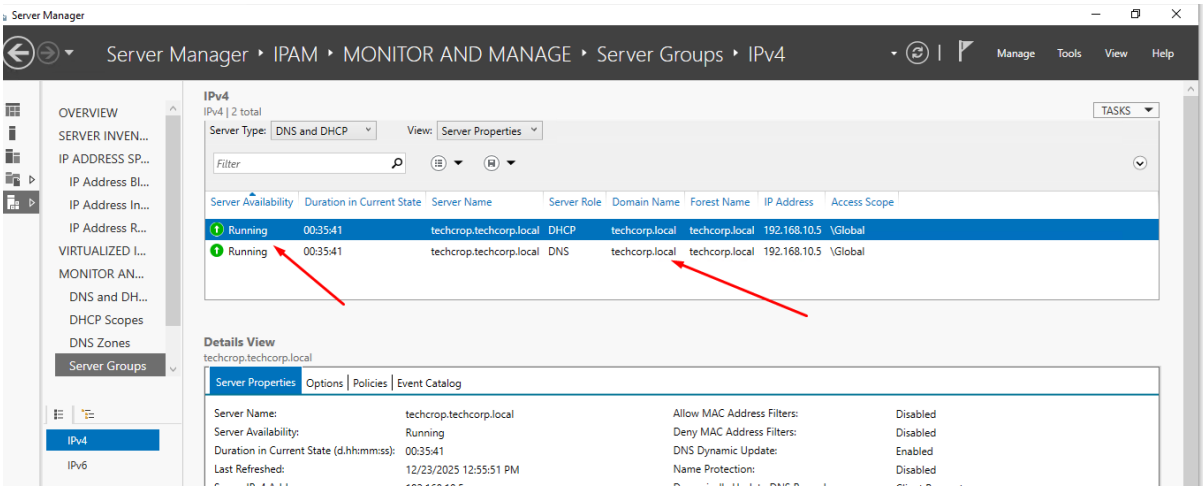
- Added the domain (techcorp.local) in **Server Discovery**.
- Started discovery to find DHCP and DNS servers.



1. **Edited GPO Access for DHCP**

- In **IPAM → Manage → Manage Servers**, selected the DHCP server.
- Changed **"Manageability Status"** to **"Managed"**.
- Granted **"Edit and view server data"** permission.





Success Verification

- In IPAM Console:
- Viewed active DHCP leases under DHCP Scopes.
- Confirmed DNS zones (`techcorp.local`, reverse zone) under **DNS Zones**.
- Verified server roles and IP address tracking.